

Best Programming Practice

1. All values as variables including Fixed, User Inputs, and Results
2. Proper naming conventions for all variables
3. Proper Program Name and Class Name
4. Proper Method Name which indicates action taking inputs and providing result

Sample Program 1: Create a program to find the sum of all the digits of a number given by a user using an array and display the sum.

- a. Use Math.random() and get a 4-digit random integer number
- b. Write a method to count digits in the number
- c. Write a method to return an array of digits from a given number.
- d. Write a method to Find the sum of the digits of the number in the array
- e. Finally, display the sum of the digits of the number

Java

```
// Create SumOfDigit Class to compute the sum of 4 digits random number
class SumOfDigits {
    // Get a 4 digit random number
    public int get4DigitRandomNumber() {
        return (int) (Math.random() * 9000) + 1000;
    }

    // Find the count of digits in the number
    public int countDigits(int number) {
        int count = 0, temp = number;
        while (temp > 0) {
            count++;
            temp /= 10;
        }
        return count;
    }

    // Store the digits of the number in an array
    public int[] getDigits(int number, int count) {
        int[] digits = new int[count];
        int temp = number;
        for (int i = count - 1; i >= 0; i--) {
            digits[i] = temp % 10;
            temp /= 10;
        }
        return digits;
    }
}
```

```
// Find the sum of the elements in an array
public int sumArray(int[] array) {
    int sum = 0;
    for (int i = 0; i < array.length; i++) {
        sum += array[i];
    }
    return sum;
}

public static void main(String[] args) {
    // Get 4 digit random integer number
    SumOfDigits sumOfDigits = new SumOfDigits();
    int number = sumOfDigits.get4DigitRandomNumber();
    System.out.println("The Random Number is: " + number);

    // Get the count of digits
    int count = sumOfDigits.countDigits(number);
    System.out.println("The count of digit is: " + count);

    // Get the array of digits from the number
    int[] digits = sumOfDigits.getDigits(number, count);

    // Find the sum of the digits of the number
    int sum = sumOfDigits.sumArray(digits);

    // Display the sum of the digits of the number
    System.out.println("\nSum of Digits: " + sum);
}
}
```

Level 1 Practice Programs

1. Write a program to input the Principal, Rate, and Time values and calculate Simple Interest.

Hint =>

- a. Simple Interest = Principal * Rate * Time / 100
- b. Take user input for principal, rate, time
- c. Write a method to calculate the simple interest given principle, rate and time as parameters
- d. Output "The Simple Interest is ____ for Principal ____, Rate of Interest ____ and Time ____"

```
import java.util.Scanner;
public class SimpleInterestCalculator {

    /*method to calculate simple interest taking principal
    rate and time as parameters pr, rate and time respectively*/
    public double simpleInterestCalc( double pr, double rate, double
time) {

        //creating variable si to indicate simple interest
        double si;

        //calculating simple interest
        si = (pr * rate * time) / 100;

        //returning the simple interest
        return si;
    }

    //main method
    public static void main(String[] args) {

        //creating scanner object
        Scanner input = new Scanner(System.in);

        /*creating variables principal, rate and time to
        indicate principal amount, rate of interest and time*/
        double principal, rate, time;

        //taking user input for principal, rate and time
        System.out.print("Enter the principal amount : ");
        principal = input.nextDouble();
```

```

        System.out.print("Enter the rate of interest : ");
        rate = input.nextDouble();
        System.out.print("Enter the time : ");
        time = input.nextDouble();

        //creating object obj for calling the method
        SimpleInterestCalculator obj = new SimpleInterestCalculator();

        /*creating variable si to store simple interest received
        by calling simpleInterestCalc method*/
        double si = obj.simpleInterestCalc(principal, rate, time);

        //printing the simple interest
        System.out.println("The Simple Interest is " + si + " for
Principal " + principal + ", Rate of Interest " + rate + " and Time " +
time);

        //closing the input object
        input.close();

    }
}

```

2. Create a program to find the maximum number of handshakes among N number of students.

Hint =>

- a. Get integer input for number of students
- b. Use the combination = $(n * (n - 1)) / 2$ formula to calculate the maximum number of possible handshakes.
- c. Write a method to use the combination formulae to calculate the number of handshakes

```

import java.util.Scanner;
public class NumberOfHandShakes2 {

    /*method to calculate maximum number of handshakes
    taking n as parameter indicating the number of persons*/
    public int handshakeCalculator( int n) {

```

```
//creating variable numberOfHandShakes to indicate
numberOfHandshakes

    int numberOfHandShakes;

    //calculating number of handshakes
    numberOfHandShakes = (n * (n - 1)) / 2;

    //returning the number of handshakes
    return numberOfHandShakes;
}

//main method
public static void main(String[] args) {

    //creating scanner object
    Scanner input = new Scanner(System.in);

    /*creating variable numberOfPersons and numberOfHandShakes
    to indicate number Of Persons, number Of HandShakes */
    int numberOfPersons, numberOfHandshakes;

    //taking number of persons as user input
    System.out.print("Enter number of persons : ");
    numberOfPersons = input.nextInt();

    //creating object obj for calling the method
    NumberOfHandShakes2 obj = new NumberOfHandShakes2();

    //calling the method
    numberOfHandshakes = obj.handshakeCalculator(numberOfPersons);

    //printing details
    System.out.println("the number of possible handshakes is " +
    numberOfHandshakes);

    //closing the input object
    input.close();

}
```

```
}
```

3. Create a program to find the maximum number of handshakes among N number of students.

Hint =>

- a. Get integer input for numberOfStudents variable.
- b. Use the combination = $(n * (n - 1)) / 2$ formula to calculate the maximum number of possible handshakes.
- c. Display the number of possible handshakes.

```
import java.util.Scanner;
public class NumberOfHandShakes2 {

    /*method to calculate maximum number of handshakes
    taking n as parameter indicating the number of persons*/
    public int handshakeCalculator( int n) {

        //creating variable numberOfHandShakes to indicate
        numberOfHandshakes
        int numberOfHandShakes;

        //calculating number of handshakes
        numberOfHandShakes = (n * (n - 1)) / 2;

        //returning the number of handshakes
        return numberOfHandShakes;
    }

    //main method
    public static void main(String[] args) {

        //creating scanner object
        Scanner input = new Scanner(System.in);

        /*creating variable numberOfPersons and numberOfHandShakes
        to indicate number Of Persons, number Of HandShakes */
        int numberOfPersons, numberOfHandshakes;

        //taking number of persons as user input
```

```

System.out.print("Enter number of persons : ");
numberOfPersons = input.nextInt();

//creating object obj for calling the method
NumberOfHandShakes2 obj = new NumberOfHandShakes2();

//calling the method
numberOfHandshakes = obj.handshakeCalculator(numberOfPersons);

//printing details
System.out.println("the number of possible handshakes is " +
numberOfHandshakes);

//closing the input object
input.close();

}
}

```

4. An athlete runs in a triangular park with sides provided as input by the user in meters. If the athlete wants to complete a 5 km run, then how many rounds must the athlete complete

Hint =>

- Take user input for 3 sides of a triangle
- The perimeter of a triangle is the addition of all sides and rounds is distance/perimeter
- Write a Method to compute the number of rounds user needs to do to complete 5km run

```

import java.util.Scanner;
public class RoundCalculator {

    /*method to calculate maximum number of handshakes
    taking n as parameter indicating the number of persons*/
    public int roundCalculation ( double side1, double side2, double
side3) {

        //creating variable numberOfRounds to indicate number Of Rounds
        int numberOfRounds;

        //calculating perimeter
        double perimeter = (side1 + side2 + side3);

```

```
//calculating rounds must be completed
numberOfRounds = 5/(int)perimeter;

//returning the number of handshakes
return numberOfRounds;
}

//main method
public static void main(String[] args) {

    //creating scanner object
    Scanner input = new Scanner(System.in);

    /*creating variables side1, side2, side3, perimeter and
    numberOfRounds to indicate 3 sides of triangle perimeter
    and number of round the athlete must complete respectively*/
    double side1, side2, side3;
    int numberOfRounds;

    //taking first side of triangle as user input
    System.out.print("Enter first side: ");
    side1 = input.nextDouble();

    //taking second side of triangle as user input
    System.out.print("Enter second side: ");
    side2 = input.nextDouble();

    //taking third side of triangle as user input
    System.out.print("Enter third side: ");
    side3 = input.nextDouble();

    //creating object obj for calling the method
    RoundCalculator obj = new RoundCalculator();

    //calling the method
    numberOfRounds = obj.roundCalculation(side1, side2, side3);

    //printing results
    System.out.print("The total number of rounds the athlete will
```



```
run is " + numberOfRounds + " to complete 5 km");

        //closing the input object
        input.close();

    }
}
```

5. Write a program to check whether a number is positive, negative, or zero.

Hint => Get integer input from the user. Write a Method to return -1 for negative number, 1 for positive number and 0 if number is zero

```
import java.util.Scanner;
public class NatureOfNumberWithMethod {

    /*method to check whether the number is positive negative or zero*/
    public int natureEvaluation(int number) {

        // Check if the number is positive, negative, or zero
        if (number > 0) {
            return 1;
        } else if (number < 0) {
            return -1;
        } else {
            return 0;
        }
    }

    //main method
    public static void main(String[] args) {

        //creating scanner object
        Scanner input = new Scanner(System.in);

        // Taking user input for the number
        System.out.print("Enter a number: ");
        int number = input.nextInt();

        //creating object obj for calling the method
        NatureOfNumberWithMethod obj = new NatureOfNumberWithMethod();
```

```
//calling the method and storing the result in result variable
int result = obj.natureEvaluation(number);

//printing results
if (result == 1) {
    System.out.println("The number is positive.");
} else if (result == -1) {
    System.out.println("The number is negative.");
} else {
    System.out.println("The number is zero.");
}

//closing the input object
input.close();

}

}
```

6. Write a program SpringSeason that takes two int values month and day from the command line and prints "Its a Spring Season" otherwise prints "Not a Spring Season".

Hint => Spring Season is from March 20 to June 20. Write a Method to check for Spring season and return a boolean true or false

```
import java.util.Scanner;
public class SpringSeasonEvaluator {

    /*method to calculate maximum number of handshakes
    taking n as parameter indicating the number of persons*/
    public void springSeason(int month, int day) {

        //creating variable numberOfRounds to indicate number Of Rounds
        int numberOfRounds;

        // Check if the date falls within the Spring Season
        if ((month == 3 && day >= 20) || (month == 6 && day <= 20) ||
(month > 3 && month < 6)) {
            System.out.println("It's a Spring Season");
        }
    }
}
```

```

        else {
            System.out.println("Not a Spring Season");
        }
    }

    //main method
    public static void main(String[] args) {

        //creating scanner object
        Scanner input = new Scanner(System.in);

        // Take user input for the month
        System.out.print("Enter the month (1-12): ");
        int month = input.nextInt();

        // Take user input for the day
        System.out.print("Enter the day (1-31): ");
        int day = input.nextInt();

        //creating object obj for calling the method
        SpringSeasonEvaluator obj = new SpringSeasonEvaluator();

        //calling the method
        obj.springSeason(month, day);

        //closing the input object
        input.close();

    }
}

```

7. Write a program to find the sum of n natural numbers using loop

Hint => Get integer input from the user. Write a Method to find the sum of n natural numbers using loop

```

import java.util.Scanner;
public class SumOfNaturalNumbers {

    /*method to calculate maximum number of handshakes
    taking n as parameter indicating the number of persons*/
    public int sumCalc (int n) {

```

```
//creating variable sum to indicate the sum
int sum=0;

//calculating sum by for loop
for(int i=1;i<=n;i++)
{
    sum += i;
}

//returning the sum
return sum;
}

//main method
public static void main(String[] args) {

    //creating scanner object
    Scanner input = new Scanner(System.in);

    /*creating variables n and sum indicating the number
    and sum of n natural numbers respectively*/
    int n, sum = 0;

    //taking user input
    System.out.print("Enter a number :");
    n = input.nextInt();

    //creating object obj for calling the method
    SumOfNaturalNumbers obj = new SumOfNaturalNumbers();

    //calling the method
    sum = obj.sumCalc(n);

    //printing results
    System.out.print("The sum of natural numbers from 1 to " + n +
    " is " + sum);

    //closing the input object
    input.close();
}
```

```
}  
}
```

8. Write a program to find the smallest and the largest of the 3 numbers.

Hint =>

- Take user input for 3 numbers
- Write a single method to find the smallest and largest of the three numbers

`public static int[] findSmallestAndLargest(int number1, int number2, int number3)`

```
import java.util.Scanner;  
public class SmallestAndLargest {  
  
    /*method to calculate maximum number of handshakes  
    taking n as parameter indicating the number of persons*/  
    public static int[] findSmallestAndLargest(int number1, int number2,  
int number3) {  
  
        // creating array result to store smallest and largest number  
        int result[] = new int[2];  
  
        //calculating smallest  
        if(number1>number2){  
            if(number2>number3){  
                result[0] = number3;  
            } else {  
                result[0] = number2;  
            }  
        } else {  
            if(number1>number3){  
                result[0] = number3;  
            } else {  
                result[0] = number1;  
            }  
        }  
  
        //calculating largest number  
        if(number1<number2){  
            if(number2<number3){  
                result[1] = number3;  
            } else {
```

```

        result[1] = number2;
    }
} else {
    if(number1<number3){
        result[1] = number3;
    } else {
        result[1] = number1;
    }
}

//returning the smallest and largest number
return result;
}

//main method
public static void main(String[] args) {

    //creating scanner object
    Scanner input = new Scanner(System.in);

    // Take user input for the first number
    System.out.print("Enter the first number: ");
    int number1 = input.nextInt();

    // Take user input for the second number
    System.out.print("Enter the second number: ");
    int number2 = input.nextInt();

    // Take user input for the third number
    System.out.print("Enter the third number: ");
    int number3 = input.nextInt();

    // creating array result to store smallest and largest number
    int result[] = new int[2];

    //calling the method
    result = findSmallestAndLargest(number1, number2, number3);

    //printing smallest and largest one
    System.out.print("The numbers are : " + number1 + " " + number2
+ " " + number3 + "\nThe smallest of three numbers is " + result[0] + " and

```

```
largest is " + result[1]);

        //closing the input object
        input.close();

    }
}
```

9. Write a program to take 2 numbers and print their quotient and remainder

Hint =>

- Take user input as integer
- Use division operator (/) for quotient and moduli operator (%) for remainder
- Write Method to find the remainder and the quotient of a number

public static int[] findRemainderAndQuotient(int number, int divisor)

```
import java.util.Scanner;
public class QuotientAndRemainderWithMethod {

    /*method to calculate maximum number of handshakes
    taking n as parameter indicating the number of persons*/
    public static int[] findRemainderAndQuotient(int number1, int
number2) {

        // creating array result to store the quotient and remainder of
numbers
        int result[] = new int[2];

        //calculating quotient and storing in first index of result
array
        result[0] = number1/number2;

        //calculating remainder and storing in the second index of
array result
        result[1] = number1%number2;

        //returning the quotient and remainder of numbers
        return result;
    }

    //main method
    public static void main(String[] args) {
```

```
//creating scanner object
Scanner input = new Scanner(System.in);

/*creating variables number1 and number2 to indicate the number
1 and number 2 respectively*/
int number1, number2, quotient, remainder;

//taking number1 and number2 as user input
System.out.print("Enter first number : ");
number1 = input.nextInt();

System.out.print("Enter second number : ");
number2 = input.nextInt();

// creating array result to store quotient and remainder of
numbers
int result[] = new int[2];

//calling the method
result = findRemainderAndQuotient(number1, number2);

//printing smallest and largest one
System.out.print("The numbers are : " + number1 + " " + number2
+ "\nThe quotient is " + result[0] + " and remainder is " + result[1]);

//closing the input object
input.close();

}
}
```

10. Create a program to divide N number of chocolates among M children. Print the number of chocolates each child will get and also the remaining chocolates

Hint =>

- Get an integer value from user for the numberOfchocolates and numberOfChildren.
- Write the method to find the number of chocolates each child gets and number of remaining chocolates

public static int[] findRemainderAndQuotient(int number, int divisor)

```
import java.util.Scanner;
```



```
public class ChocolateDistribution2 {

    /*method to calculate maximum number of handshakes
    taking n as parameter indicating the number of persons*/
    public static int[] chocolate(int numberOfchocolate, int
numberOfChildren) {

        // creating array result to store the quotient and remainder of
numbers
        int result[] = new int[2];

        //calculating number of chocolate each child get
        result[0] = numberOfchocolate/numberOfChildren;

        //calculating remaining chocolates
        result[1] = numberOfchocolate%numberOfChildren;

        //returning the result
        return result;
    }

    //main method
    public static void main(String[] args) {

        //creating scanner object
        Scanner input = new Scanner(System.in);

        /*creating variables numberOfchocolate, numberOfChildren,
        to indicate the number of chocolate and number of children
        respectively*/
        int numberOfchocolate, numberOfChildren;

        //taking number of chocolate and number of children as user
input
        System.out.print("Enter number of chocolate : ");
        numberOfchocolate = input.nextInt();

        System.out.print("Enter number of children: ");
        numberOfChildren = input.nextInt();
    }
}
```

```

/* creating array result to store number of chocolate
each child get and remaining chocolate*/
int result[] = new int[2];

//calling the method
result = chocolate(numberOfchocolate, numberOfChildren);

//printing details
System.out.println("The number of chocolates each child gets is
" + result[0] + " and the number of remaining chocolates are " +
result[1]);

//closing the input object
input.close();

}
}

```

11. Write a program calculate the wind chill temperature given the temperature and wind speed

Hint =>

- Write a method to calculate the wind chill temperature using the formula

$$windChill = 35.74 + 0.6215 * temp + (0.4275 * temp - 35.75) * windSpeed^{0.16}$$

public double calculateWindChill(double temperature, double windSpeed)

```

import java.lang.Math;
import java.util.Scanner;
public class Wind {

    /*method to calculate maximum number of handshakes
    taking n as parameter indicating the number of persons*/
    public static double windChillCalc(double temperature, double
windSpeed) {

        //creating variable numberOfRounds to indicate number Of Rounds
        double windChill;

        //calculating wind chill
        windChill = 35.74 + 0.6215 * temperature + (0.4275 *
temperature - 35.75) * Math.pow(windSpeed, 0.16);
    }
}

```

```

        //returning wind chill
        return windChill;
    }

    //main method
    public static void main(String[] args) {

        //creating scanner object
        Scanner input = new Scanner(System.in);

        /*creating variables temperature, windSpeed and windChill to
        indicate the
        temperature, wind speed and windchill respectively*/
        double temperature, windSpeed, windChill;

        //taking temperature and windChill as user input
        System.out.print("Enter temperature : ");
        temperature = input.nextDouble();

        System.out.print("Enter windChill : ");
        windSpeed = input.nextDouble();

        //calling the method
        windChill = windChillCalc(temperature, windSpeed);

        //printing details
        System.out.print("Temperature = " + temperature + "\nWind Speed
= " + windSpeed + "\nWind chill = " + windChill);

        //closing the input object
        input.close();

    }
}

```

12. Write a program to calculate various trigonometric functions using Math class given an angle in degrees

Hint =>

- a. Method to calculate various trigonometric functions, Firstly convert to radians and then use Math function to find sine, cosine and tangent.

public double[] calculateTrigonometricFunctions(double angle)

```
import java.lang.Math;
import java.util.Scanner;
public class Trigonometry {

    //method to calculate sine cosine and tangent of angle
    public static void trigonometryFunc( double angle) {

        //creating variables sin, cos and tan to store sine, cosine and
        tangent of angle
        double sin, cos, tan;
        //calculating sine
        sin = Math.sin(angle);

        //calculating cosine
        cos = Math.cos(angle);

        //calculating tangent
        tan = Math.tan(angle);

        //printing details
        System.out.println("Angle = " + angle + "\nsine = " + sin +
        "\ncosine = " + cos + "\ntangent = " + tan);
    }

    public static void main(String[] args) {

        //creating scanner object
        Scanner input = new Scanner(System.in);

        //creating variable angle
        double angle;

        //taking user input
        System.out.print("Enter the angle : ");
        angle = input.nextDouble();

        //calling the method
        trigonometryFunc(angle);
    }
}
```

```
//closing the scanner object  
input.close();  
  
    }  
}
```